Yi Hu

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Personal Summary

PhD researcher specialized in **embodied AI** and **surgical robotics**. Developed **Vision-Language-Action (VLA) model** and **Reinforcement Learning (RL)/Imitation Learning (IL)** frameworks for Franka and da Vinci surgical robots. Strong background in SLAM, kinematics/dynamics, and control theory.

Education

PhD in Biomedical Engineer — Jan. 2021 – Exp. Apr. 2026
University of Alberta — *Edmonton, Canada*

MSc in Control Theory and Control Engineering — Sept. 2017 – Jun. 2020
Sichuan University — *Chengdu, China*

B.S. in Process Equipment and Control Engineering — Sept. 2013 – Jun. 2017
Hefei University of Technology — *Hefei, China*

Project Experience

Human-Centered Autonomy Lab — Jan. 2024 – Present
Graduate Research Assistant, University of Alberta — *Edmonton, Canada*

- Built a large **VLA** model with **3D spatial reconstruction** and **mixture-of-expert (MoE)** model, and validated in robotic simulator environment (Libero).
- **VLA workflow: Collected** and **post-training** pi0.5 model on large-scale multimodal datasets (**over 300G, 5 manipulation tasks**), **deployed** for real-time **RGB-D inference** on a Franka robot.
- Engineered a **real-time motion policy** for dynamic **fast manipulation** by architecting a single-step **diffusion model**; Optimized the inference engine to achieve 7-10x lower latency; Demonstrated robustness via **sim2real** transfer (**PyBullet** to **Franka robot**)
- Developed a **Reinforcement Learning (RL) framework** transferring **human datasets** priors to robotic surgical subtasks. Enabled optimized **patient side manipulation** and **endoscopic camera operation** from human daily demonstrations. Demonstrated robustness via **sim2real** transfer (**PyBullet** to **da Vinci surgical robot**)

Telerobotic and Biorobotic Systems Group — Jan. 2021 – Dec. 2023
Graduate Research Assistant, University of Alberta — *Edmonton, Canada*

- Built an **end-to-end** autonomous ultrasound system on a Franka robot. Fused **classical control** (admittance control) with a real-time **deep learning image processing** to regulate probe contact. Engineered data foundation that collects and **auto-labels multimodal sensor data** from **force-torque sensor** and **video grabber**.
- Developed an autonomous endoscopic camera system for the **da Vinci surgical robot** by **Imitation Learning (IL)** from noisy surgical data; Engineered a **safety-critical** and **collision-aware motion planning** framework .
- Developed a **real-time teleoperation** pipeline that transformed **free-hand movements** into **6-DoF robot** control; Integrated a lightweight **gesture classifier** trigger discrete robot actions.

Perception, Motion Control and Intelligent Robot Innovation Lab — Sept. 2017 – Dec. 2020
Graduate Research Assistant, Sichuan University — *Chengdu, China*

- Implemented a real-time **event-camera SLAM** pipeline for **mobile robot navigation** with **scene understanding**, enabling 6-DoF **motion tracking** and **3D reconstruction** through optimization-based mapping and keyframe back-end refinement. Built a **physics-based modeling** in Unity for mobile robot navigation.
- Designed robust **nonlinear control** algorithms for **joint-space control**, accounting for **actuator saturation** and incorporated **disturbance observers**, with **system stability** guaranteed through Lyapunov analysis.
- Derived **kinematic and dynamic models** for a wheel mobile robot, enabling motion planning, control design, and trajectory tracking.

Skills

Programming and Frameworks: Python, MATLAB/Simulink, C++, Keras, TensorFlow, Pytorch, Ubuntu
Robotics and Tools: MuJoCo, Isaac Gym, PyBullet, ROS2, RViz, Gazebo, Git, SolidWorks, AutoCAD, Ansys, Comsol, LaTeX
Soft Skills: Communicating cross functionally, Continuous Learning, Innovative Thinking, Event Management
Languages: Mandarin(Native), Cantonese(Native), English(Fluent)